

Gym and Fitness Membership Management System

Database Systems Project Proposal

List Of Members:

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1- Introduction and Brief Description of the System

- a. The Gym & Fitness Membership Management System is a web-based application designed to efficiently manage the operations of a gym, including member registration, trainer management, session scheduling, workout tracking, and payment handling. In traditional gym environments, most processes such as maintaining member records, tracking attendance, and managing payments are handled manually, which can lead to errors, inefficiencies, and data redundancy. This system aims to automate these processes and provide a centralized platform for managing all gym-related activities. The system will allow administrators to manage members, trainers, and membership plans; trainers to manage sessions and workouts; and members to track their fitness progress, attendance, and payments. Overall, the system improves operational efficiency, accuracy, and user experience.

2- System Functionality

a. Member Management

- i. Registering new members
- ii. Maintaining member profiles
- iii. Tracking membership status and history

b. Trainer Management

- i. Add and manage trainers
- ii. Assign trainers to sessions
- iii. Track trainer schedules

c. Membership Plan Management

- i. Create and manage membership plans
- ii. Define duration and pricing
- iii. Assign plans to members

d. Session Scheduling

- i. Create training sessions
- ii. Allow members to book sessions
- iii. Manage session capacity

e. Workout Tracking

- i. Create workout routines
- ii. Record workouts performed by members
- iii. Track fitness progress

f. Attendance Management

- i. Record daily attendance of members
- ii. Maintain attendance history

g. Payment Management

- i. Record membership payments
- ii. Track payment history
- iii. Monitor payment status

h. Equipment Management

- i. Maintain records of gym equipment
- ii. Track maintenance schedules

3- Users of the System

a. Admin (has full control over the system)

- i. Manage members and trainers
- ii. Create and update membership plans
- iii. Assign trainers to sessions
- iv. Manage equipment records
- v. Monitor attendance and payments
- vi. Generate reports

b. Trainer (responsible for managing workouts and sessions)

- i. Create and manage training sessions
- ii. Assign workouts to members
- iii. Record workout performance
- iv. Track member progress
- v. Mark attendance

c. Member (uses the system to manage their fitness activities)

- i. View membership details
- ii. Book training sessions
- iii. Track workout history
- iv. View attendance records
- v. Check payment history

4- Technologies Used

a. Front-end

- i. HTML
- ii. CSS
- iii. JavaScript (React.js)
- iv. Bootstrap

b. Back-end

- i. Python

c. Database

- i. Postgresql

5- Scope of the System

a. Included

- i. Member registration and management
- ii. Trainer management
- iii. Membership plan management
- iv. Session scheduling and booking
- v. Workout tracking and logging
- vi. Attendance management
- vii. Payment recording and tracking
- viii. Equipment management

b. Not Included

- i. Online payment gateway integration
- ii. Mobile application development
- iii. AI based recommendations
- iv. Integration with wearable fitness devices

6- Assumptions

- a. Each member can have only one active membership plan at a time.
- b. Trainers can conduct multiple training sessions.
- c. Members can attend multiple sessions based on their membership plan.
- d. Payments are recorded manually by the administrator.
- e. All users have valid login credentials to access the system.